CHEMISTRY ASSIGNMENT 28

In this assignment, you must show that you have used a variety of sources. Include information from textbooks, periodicals, CD ROMs, the Internet and encyclopaedias.

Don't forget to include a bibliography!

1 The scientists listed below all worked in chemistry. Describe briefly (five or six sentences each) the major contributions of each scientist. Include where and when their research was carried out.

- a James Chadwick
- b Neils Böhr
- c John Dalton
- d JJ Thomson
- e Ernest Rutherford
- f Dmitri Menedeléev (Note that spellings vary.)
- g Amadeo Avogadro
- h Gay Lussac
- i Democritus
- j Robert Bunsen

2 Name one person involved in chemistry who was not mentioned in the previous question. Outline his or her contribution to chemistry.

3 Give a brief definition (two or three lines) for each term.

- a ion
- b sub-atomic particle
- c molecule
- d isotope
- e Periodic Table
- f polymer
- g organic
- h acid
- i base
- j atom
- k alcohol/alkanol
- l pH
- m octet
- n mole

4 Chemistry is one of the categories for the Nobel Prize. Find out:

- a what the Nobel Prize is
- b how it began
- c why it is awarded
- d the name of a scientist or scientific team who was awarded the Nobel Prize in chemistry and the year the prize was awarded.

5 The Kelvin is a scale of temperature. Define the Kelvin in chemistry terms and give a brief description of the scientist after whom it was named.

6 A law is a statement that is considered to be true. There are basic laws in chemistry. These include the *law of conservation of mass* and the *law of definite proportions* (sometimes called the *law of constant composition*). Give brief definitions (one or two lines) for these two laws.

7 Match each term to the correct definition.

- a titration
- b polar
- c energy
- d binary

- i means two (containing two elements)
- ii the ability of a system to do work
- iii attraction between oppositely charged particles
- iv process to determine concentration of a solution

8 Chemists are employed in many areas of society, such as education, industry, medicine and government institutions. Research one area in which chemists contribute to our way of life. In your answer, state the precise area of chemistry and decribe what chemists in this particular occupation do.

9 Explain the difference between a chemist and a pharmacist.

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